

Bimodal Passenger Flow Data Reporting Protocol

preface

Passenger Flow Data Protocol Description

Data reporting uses the HTTP protocol with POST+JSON method.

Functions: Heartbeat, Data Upload.

Heartbeat mechanism: Upload heartbeat data every minute, including SN, timestamp, and status.

Upload frequency: Set to 0 for real-time transmission, set to 1 for once every minute, set to 5 for once every 5 minutes, and set to 60 for once every 60 minutes

Data upload content: SN, data timestamp, start time of statistics, end time of statistics, number of entries, number of exits, number of passes.

Data upload mode: Passenger flow data supports incremental and aggregated reporting (defaulting to incremental reporting if no response is received)

Command to retrieve historical data: Retrieves historical data within the past 90 days

Reid protocol specification

Data reporting uses the HTTP protocol with POST+JSON method.

Functions: Real-time deduplication, deduplicated data, and Reid data upload.

Working Principle: The device collects data in real time, processes it after business hours, and uploads the results.

Upload frequency: Once daily after data analysis is complete at the end of working hours

Real-time deduplication and statistics; reports are submitted immediately after matching (the number of matched users may fluctuate).

Passenger Flow Data Reporting

1. Heartbeat Interface

Device Request Method: POST

Server address example: `http://https://XXXX.XXXX.XXXX.XXXX:XXApi/camera/heartBeat`

The server path follows the recommended format for easier configuration

Heartbeat Reporting Parameter Format

The reported data is in JSON format with the following structure:

```
{
  "macAddress": "4C:BC:98:66:00:5E",
  "ipAddress": "0.0.0.0",
  "connectionType": "Wifi",
  "wifiSSID": "hongwaiceshi",
  "wifiPassword": "Hxzy87996629.",
  "ipAddressMethod": "DHCP",
  "hostName": "CAM-0",
  "timeZone": 8,
  "hwPlatform": "V5.0",
  "swRelease": "V7.2.3.9-V5-21-RT",
  "sn": "201000002501090095",
  "timestamp": 1752819461
}
```

Field Description

field name	required	type	explain
sn	yes	string	Device Serial Number
timestamp	yes	long	Unix timestamp for time synchronization
macAddress	No	string	Device MAC Address (pushes when property information is enabled)
ipAddress	No	string	Device IP Address (pushes when the property information i

			s enabled)
connectionType	No	string	Connection Type [Wired or Wi-Fi] (push notifications when enabled)
hostName	No	string	Current device host name (pushed when attribute information is enabled)
wifiSSID	No	string	SSID for WiFi connection (valid only when connected to Wi-Fi) (shown after enabling property settings)
wifiPassword	No	string	WiFi connection password (valid only when connecting to Wi-Fi) (will be displayed after enabling property information)
ipAddressMethod	No	string	IP address acquisition method [Static or DHCP] (pushes information after enabling properties)
hwPlatform	No	string	Hardware Version (push notification enabled when property information is activated)
swRelease	No	string	Software Version (push notification enabled when property information is activated)
timeZone	No	string	Time Zone (e.g., 8 represents Eastern Time Zone 8) (push notifications when enabled)

Heartbeat Interface Server Response Description

The server returns the following content structure:

Success Example:

```
{
  "msg": "success",
  "code": 0,
  "data": {
    "uploadInterval": 0,
    "dataMode": "Add",
    "timezone": 8,
    "dataEndTime": 1752854340,
    "sn": "201000002501090095",
    "time": 1752820142,
    "dataStartTime": 1752768000
  }
}
```

Failure example:

```
{
  "code": 1,
  "msg": "Returns information describing an error;"
}
```

Field Description

field name	required	type	explain
code	yes	int	Return code: 0: Success; 1: SN does not exist; 2: Other: Failure
msg	yes	string	return information
data.sn	yes	string	Device Serial Number
data.dataStartTime	No	long	Unix timestamp: when the synchronized device began reporting
data.dataEndTime	No	long	Unix timestamp indicating the end of reporting by synchronized devices
data.uploadInterval	No	long	Data Reporting Interval (minutes)
data.dataMode	No	string	"Add" is incremental, while "Total" is cumulative
data.time	yes	long	Unix timestamp, server time, used for time synchronization
data.timezone	No	long	GMT time zone

2. Data Upload Interface

Device Request Method: POST

Server address: <http://XXXX.XXXX.XXXX.XXXX:XX/api/camera/dataUpload>

The server path follows the recommended format for easier configuration

Data Reporting Parameter Format

The reported data is in JSON format with the following structure:

```
{
  "macAddress": "4C:BC:98:66:00:5E",
  "ipAddress": "0.0.0.0",
  "connectionType": "Wifi",
  "wifiSSID": "hongwaiceshi",
  "wifiPassword": "Hxzy87996629.",
  "ipAddressMethod": "DHCP",
  "hostName": "CAM-0",
  "timeZone": 8,
  "hwPlatform": "V5.0",
  "swRelease": "V7.2.3.9-V5-21-RT",
}
```

```

"sn": "201000002501090095",
"time": 1752820828,
"startTime": 1752820828,
"endTime": 1752820828,
"in": 1,
"out": 0,
"passby": 0,
"turnback": 0,
"avgStayTime": 1640,
"len": 1,
"attributes":
[
{
"idIndex": 2458,
"personId": 2458,
"timeStamp": 1752820828000000,
"eventType": 0,
"stayTime": 1640,
"lineId": 0,
"height": 164,
"gender": 1,
"age": [31, 45],
"workcard": 0,
"wheelchair": 0
}
]
}

```

Field Description

field name	required	type	explain
sn	yes	string	Device Serial Number
startTime	yes	long	Unix timestamp, start time for statistics
endTime	yes	long	Unix timestamp, end time of statistics
time	yes	long	Unix timestamp, data time
in	yes	int	Inbound passenger flow
out	yes	int	Outbound Passenger Flow
passby	yes	int	Number of people
turnback	yes	int	Number of returnees

avgStayTime	yes	int	Average dwell time within the camera field of view (unit: ms)	
macAddress	No	string	Device MAC Address	
ipAddress	No	string	Device IP Address	
connectionType	No	string	Connection Type [Wired or Wi-Fi]	
hostName	No	string	Current Device Host Name	
wifiSSID	No	string	SSID for WiFi connection (valid only when connected to WiFi)	
wifiPassword	No	string	WIFI connection password (valid only when connecting to WiFi)	
ipAddressMethod	No	string	IP Address Acquisition Method [Static or DHCP]	
hwPlatform	No	string	Hardware Version	
swRelease	No	string	software release	
timeZone	No	long	Time Zone (e.g., 8 represents Eastern Time Zone 8)	
len	No	string	Number of Data Records	
attributes	No	string	Human Body Attribute Array	
attributes.idIndex	No	INT	event ID	
attributes.personId	No	INT	personnel ID	
attributes.timeStamp	No	INT	Data Timestamp	
attributes.eventType	No	INT	Event Type [0: Enter, 1: Leave, 2: Pass Through, 3: Return]	
attributes.stayTime	No	INT	residence time	
attributes.lineId	No	INT	No meaning	
attributes.height	No	INT	Personnel Height	
attributes.age	No	short	Lower Age Limit 255 represents invalid	age <10 (Children) 10–16 (adolescents) 17~30 31~45 46~60 >60 Invalid gender; regardless of age, it is invalid
		short	Age Range Upper Limit 255 represents invalid	
attributes.gender	No	INT	sex	

			-1: Invalid 0: Undefined 1: Male 2: Female Invalid age; regardless of gender, it is invalid
attributes.wheelchair	No	INT	Wheelchair testing: 0: No wheelchair detected 1: Wheelchair detected
attributes.workcard	No	INT	ID Card Detection: 0: No ID badge detected 1: Employee ID card detected

Data Interface Server Response Description

The server returns the following content structure:

Success Example:

```
{
  "code":0,
  "msg": "Report submitted successfully"
  "data":{
    "sn":"2010010000000000"
    "time":1648606824
  }
}
```

Failure example:

```
{
  "code": 1,
  "msg": "Returns information describing an error;"
}
```

Field Description

field name	required	type	explain
code	Yes	int	Return code: 0: Success; 1: SN does not exist; 2: Other: Failure
msg	Yes	string	return information
data.sn	Yes	string	Device Serial Number
data.time	No	long	Unix timestamp, server time, used to synchronize device time

3、Residence Data Push (This option is optional; integrate it when needed)

Device Request Method: POST

Server address: `http://XXXX.XXXX.XXXX.XXXX:XX/api/camera/dataUpload`

The server path follows the recommended format for easier configuration

Residence Data Reporting Format

The request data is in JSON format with the following structure:

```
{
  "sn": "201000002209190025",
  "time": 1667802959,
  "currentStay": 2,
  "info": [
    {
      "id": 1,
      "height": 170,
      "status": 1,
      "staytime": 2000,
    },
    {
      "id": 2,
      "height": 171,
      "status": 1,
      "staytime": 3000,
    }
  ]
}
```

Field Description

field name	required	type	explain
currentStay	Yes	long	Current number of residents

info.personId	Yes	long	The internal ID for resident personnel is unique and reset daily (IDs not validly identified are not uploaded)
info.height	Yes	int	Personnel Height
info.status	Yes	int	Camera internal information and presence status may not be parsed by the platform
info.staytime	Yes	int	Current ID holder's stay duration

REID Data Reporting Interface

1. Reid Data Interface

Device Request Method: POST

Server address example: <http://> or <https://>XXXX.XXXX.XXXX.XXXX:XX/reid

The server path can be customized

Reid (residence duration) data constitutes a complete personnel ReID record, containing each physical individual's global ID, entry/exit pairing times, and physiological attributes, which is transmitted after business hours conclude.

Reid Data Reporting Format

The request data is in JSON format with the following structure:

```
{
  "date": "2026-03-04",
  "direction": "in",
  "master_sn": "CAM-A01",
  "device_sns": ["CAM-A01", "CAM-B02"],
  "reid_records": [
    {
      "global_id": 1001,
      "pairs": [
        {
          "enter": {
            "camera_sn": "CAM-A01",
            "person_id": 3,
            "timestamp_ms": 1741046400000,
            "image_path": "CAM-A01_3.jpg"
          },
          "leave": {
            "camera_sn": "CAM-A01",
```

```

    "person_id": 3,
    "timestamp_ms": 1741047300000,
    "image_path": "CAM-A01_3.jpg"
  },
  "dwell_time_ms": 900000
}
],
"extra_info": {
  "age_range_min": 17,
  "age_range_max": 30,
  "gender": 1,
  "person_type": 0,
  "height_category": 1
}
}
]
}

```

Reid Field Description

■ Top Field

field name	required	type	explain
date	Yes	string	Data Date, Format YYYY-MM-DD
direction	Yes	string	Direction: "in" (into) or "out" (out of)
master_sn	Yes	string	Main Device Serial Number (Camera SN to be aggregated)
device_sns	Yes	array	List of all device SNs involved in the statistics
reid_records	Yes	array	ReID record list, where each element represents a unique identifier

■ reid_records[] -ReID Record

Each element represents a unique, device-wide identifier that includes all of the person's in/out pairs and physical attributes.

field name	required	type	explain
global_id	Yes	long	Global Unique Identity (allocated after deduplication)

			across devices)
pairs	Yes	array	Enter and exit the pairing list; see the pairs[] field description.
extra_info	No	object	Additional human attribute information; see the description of the extra_info field

■ reid_records[]. pairs [] -Entry and exit pairs

Each pair represents a complete entry and exit process of a person within a camera's field of view (entry → exit).

A person may have multiple entries and exits on the same day, generating one pair each time; or may generate one pair under different cameras.

field name	required	type	explain
enter	Yes	object	Enter the event object and refer to the event object field descriptions
leave	Yes	object	Exit the event object and refer to the event object field description
dwel_time_ms	Yes	long	Residence duration in milliseconds (leavetimestamp_ms-entertimestamp_ms)

■ Event object (same enter/leave structure)

field name	required	type	explain
camera_sn	Yes	string	Camera serial number that captured this event
person_id	Yes	long	Person ID within a single camera
timestamp_ms	Yes	long	Event timestamp, millisecond-level Unix timestamp
image_path	No	string	Corresponding image file name

■ reid_records[].extra_info — Additional information

Includes the person's physical attributes and identity type.

field name	required	type	explain
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age_range_min	No	int	Lower age limit (see enumeration value definition)
age_range_max	No	int	Upper age limit (see enumeration value definition)
gender	No	int	Gender (see enumeration value definition)
person_type	No	int	Person Type (see enumeration values definition)
height_category	No	int	Adult/Child Classification (see enumeration value definitions)

■ Enumeration Value Definition

extra_info.age_range_min / age_range_max — Age range

price	meaning
0 / 9	0–9 years old
10 / 16	10–16 years old
17 / 30	17–30 years old
31 / 45	31–45 years old
46 / 60	Ages 46–60
61 / 99	61 years of age or older
-1 / -1	unknown

extra_info.gender-Gender

price	meaning
1	man (Male)
2	Female
0	Unknown

extra_info(person_type — Person Type

price	meaning
-------	---------

0	Customer
1	Staff
2	Rider
3	Courier

extra_info.height_category — Adult/Child Category

price	meaning
1	Adult
2	Child
0	Unknown

Server Response Parameter Description

field name	required	type	explain
code	Yes	int	Return code: 0 – Successful; 1 – SN does not exist; 2 – Other failures
msg	Yes	string	return information

Example response result:

Success Example:

```
{
  "code":0,
  msg: "Report submitted successfully"
}
```

Failure example:

```
{
  "code": 1,
  msg: "sn does not exist"
}
```

Remove duplicate data reporting interface

1. Unique Data Interface

Device Request Method: POST

Server address example: `http://` or `https://XXXX.XXXX.XXXX.XXXX:XX/dup`

The server path can be customized

Duped data refers to deduplicated passenger flow data, divided into two categories:

Real-time updated data: Includes only aggregated passenger flow statistics, sent periodically during business hours.

Final DUP Data: Add a "records" field (containing the original event list for each physical person) to the real-time DUP, and send it after business hours end.

Real-time Data Format

Contains only deduplicated summary statistics, excluding the records[] field:

```
{
  "date": "2026-03-04",
  "direction": "in",
  "master_sn": "CAM-A01",
  "device_sns": ["CAM-A01", "CAM-B02"],
  "dedup_summary": {
    "raw": 320,
    "duplicate": 85,
    "deduped": 235
  },
  "deduped_stats": {
    "customer": {
      "count": 174,
      "height_category": {"adult": 140, "child": 28, "unknown": 6},
      "age_gender": {
        "0_9": {"male": 1, "female": 2, "unknown": 0},
        "10_16": {"male": 5, "female": 6, "unknown": 1},
        "17_30": {"male": 28, "female": 27, "unknown": 3},
        "31_45": {"male": 30, "female": 27, "unknown": 2},
        "46_60": {"male": 20, "female": 14, "unknown": 0},
        "61_plus": {"male": 4, "female": 2, "unknown": 0},
        "unknown": {"male": 1, "female": 0, "unknown": 1}
      }
    }
  },
  "non_customer": {
```

```

"count": 61,
"categories": {"staff": 40, "rider": 15, "courier": 6}
}
}
}

```

Final DUP Data Format

Based on real-time DUP data, add a new "records" field:

```

{
  "date": "2026-03-04",
  "direction": "in",
  "master_sn": "CAM-A01",
  "device_sns": ["CAM-A01", "CAM-B02"],
  "dedup_summary": { "raw": 320, "duplicate": 85, "deduped": 235 },
  "deduped_stats": {
    "customer": {
      "count": 174,
      "height_category": { "adult": 140, "child": 28, "unknown": 6 },
      "age_gender": { "0_9": { "male": 1, "female": 2 }, ... }
    },
    "non_customer": {
      "count": 61,
      "categories": {"staff": 40, "rider": 15, "courier": 6}
    }
  },
  "records": [
    {
      "global_id": 0,
      "events": [
        { "camera_sn": "CAM-A01", "person_id": 42, "timestamp_ms": 1709539200000, "image_path": "CAM-A01_42.jpg" },
        { "camera_sn": "CAM-B02", "person_id": 17, "timestamp_ms": 1709539205000, "image_path": "CAM-B02_17.jpg" }
      ],
      "extra_info": {
        "age_range_min": 17,
        "age_range_max": 30,
        "gender": 2,
        "person_type": 0,
        "height_category": 1
      }
    }
  ]
}

```

}

Field Description

■ Top-level field (included in both real-time DUP and final DUP)

field name	required	type	explain
date	Yes	string	Data Date, Format YYYY-MM-DD
direction	Yes	string	Direction: "in" (into) or "out" (out of)
master_sn	Yes	string	Main Device Serial Number
device_sns	Yes	array	List of all device SNs involved in the statistics
dedup_summary	Yes	object	Remove duplicate summaries; see the description of the dedup_summary field
deduped_stats	Yes	object	After deduplication, view statistics by personnel type in the deduped_stats field description.
records	No	array	[Only Final DUP] Original event list for each physical person

■ dedup_summary-Deduplicate and summarize

field name	required	type	explain
raw	Yes	int	Total number of original events (before deduplication)
duplicate	Yes	int	Number of Repeated Events
deduped	Yes	int	Unique count after deduplication (= raw-duplicate)

■ deduped_stats — Statistics after deduplication

After deduplication, individuals are categorized into two groups: customers (customer) and non-customers (non_customer), which are mutually exclusive and fully cover the total number of deduplicated entries.

deduped_stats/customer — Customer statistics

field name	required	type	explain
count	Yes	int	Total Number of Customers
height_category	Yes	object	By height: adult / child / unknown
age_gender	Yes	object	Cross-tabulation by age group × gender; see the explanation below

Age/Gender | Age Range | Field Name Comparison:

key name	meaning
"0_9"	0–9 years old
"10_16"	10–16 years old
"17_30"	17–30 years old
"31_45"	31–45 years old
"46_60"	Ages 46–60
"61_plus"	61 years of age or older
"unknown"	Age unknown

Each age group includes a gender field: male, female, or unknown

deduped_stats.non_customer — Non-customer statistics

Non-customers refer to individuals not included in passenger flow analysis (such as employees, delivery riders, and couriers).

field name	required	type	explain
count	Yes	int	Total Number of Non-Customers
categories	Yes	object	Count by identity type: staff / rider / courier

■ Records [] -Original event list for individual entities (only the final DUP)

field name	required	type	explain
------------	----------	------	---------

global_id	Yes	long	Global Unique Identity ID
events	Yes	array	Identify the event list; see the events[] field description
extra_info	No	object	Add additional human attributes and identity types; see the description of the extra_info field.

Records[]. Events[] — Identifies events

The same global_id may be captured multiple times by different cameras at different times, with each capture recorded as an event.

field name	required	type	explain
camera_sn	Yes	string	Capture the camera serial number
person_id	Yes	long	Person ID within a single camera
timestamp_ms	Yes	long	Event Timestamp (millisecond-level Unix)
image_path	Yes	string	Corresponding image file name

arrays[].extra_info — Additional information (enum values as defined in the REID section)

field name	required	type	explain
age_range_min	No	int	Lower age limit (enum values same as in the REID section)
age_range_max	No	int	Upper age limit (enum values same as in the REID section)
gender	No	int	Gender (enum values same as in the REID section)
person_type	No	int	Person Type (values same as in the REID section)
height_category	No	int	Adult/Child Category (values same as in the REID section)

Response Parameter Description

field name	required	type	explain
code	Yes	int	Return code: 0 – Successful; 1 – SN does not exist; 2 – Other failures
msg	Yes	string	return information

Example response result:

Success Example:

```
{  
  "code":0,  
  msg: "Report submitted successfully"  
}
```

Failure example:

```
{  
  "code": 1,  
  msg: "sn does not exist"  
}
```

Version: V2.5 Date: April 24,2026

Revised REID data reporting interface (with new fields including extra_info, master_sn, device_sns, dwell_time_ms, etc.)

Update the deduplication data reporting interface (separate real-time and final DUP formats, add fields such as deduped_stats and records)